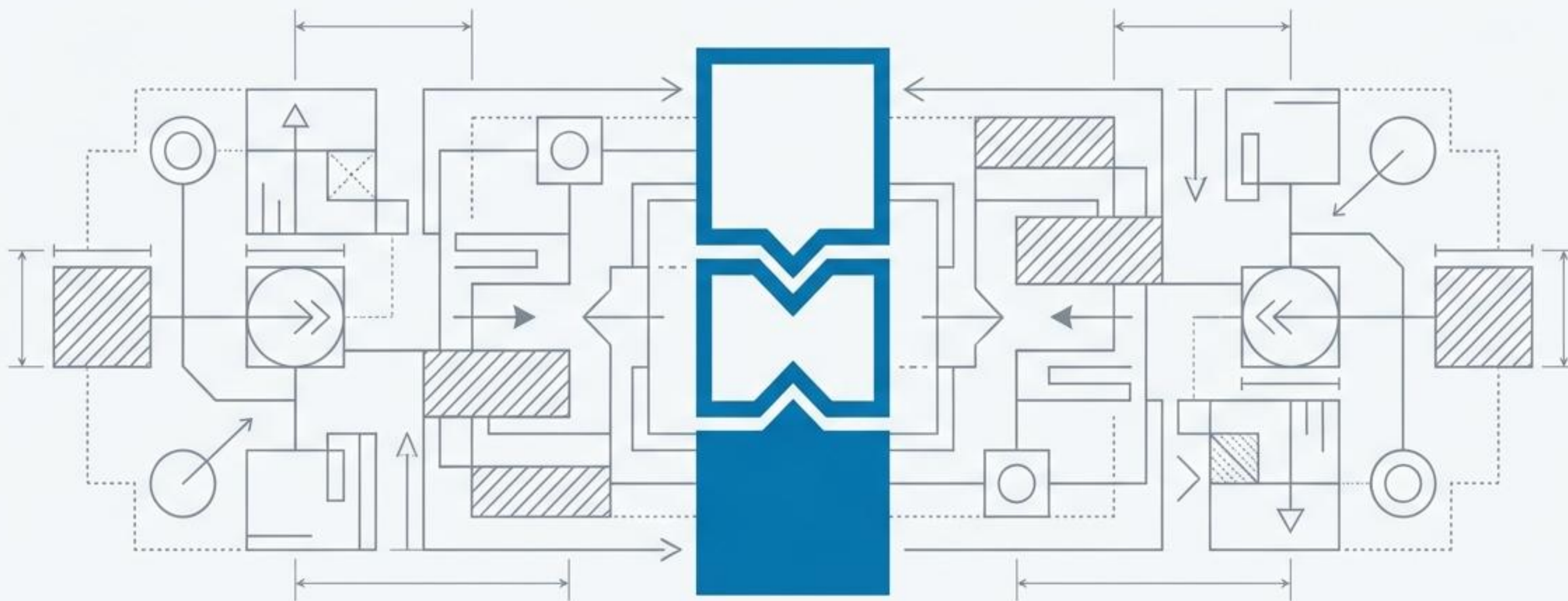


The Quant's Execution Toolkit

A Structured Guide to Order Types and Algorithms

By Michael Fernandes



A TRADER'S TOOLKIT: CHOOSING YOUR EXECUTION STRATEGY

Different order types offer a fundamental trade-off: market orders guarantee execution but sacrifice price control, while limit orders guarantee price but sacrifice execution certainty. Advanced orders and algorithms build on this to automate strategies and minimise the market impact of large trades.

FOUNDATIONAL ORDER TYPES

MARKET ORDER: Certainty Over Price



Executes immediately at the best available price, guaranteeing the trade is filled.

	PRICE CONTROL	EXECUTION CERTAINTY
MARKET ORDER	Low	High (Guaranteed)

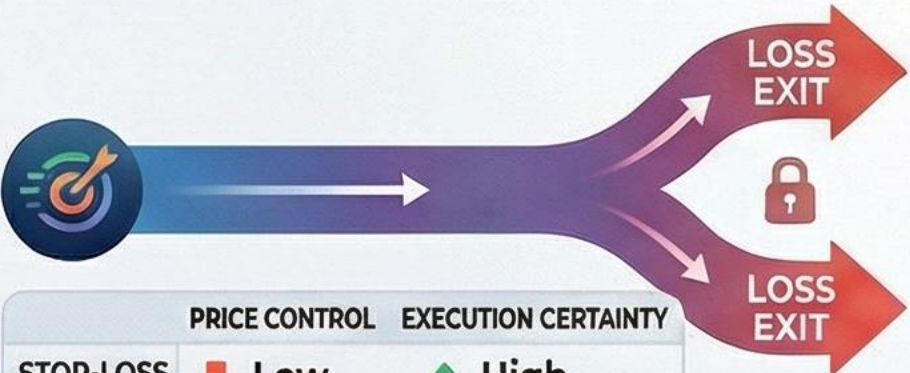


LIMIT ORDER: Price Over Certainty

Executes only at your specified price or better, but may not fill.

	PRICE CONTROL	EXECUTION CERTAINTY
MARKET ORDER	↑ High	Low (Not Guaranteed)

	PRICE CONTROL	EXECUTION CERTAINTY
LIMIT ORDER	↑ High	↓ Low (Not Guaranteed)

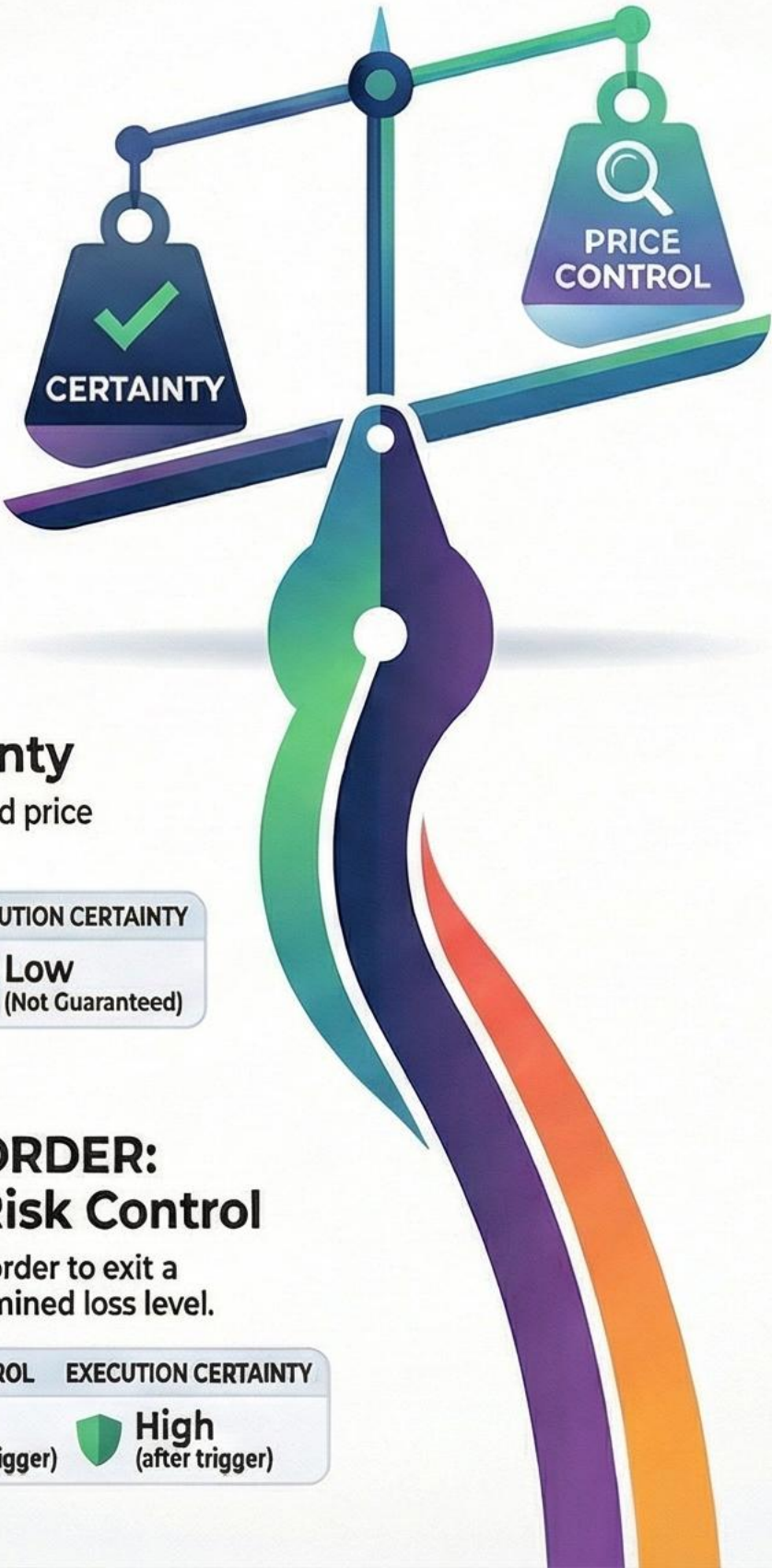


STOP-LOSS ORDER: Automated Risk Control

Converts to a market order to exit a position at a predetermined loss level.

	PRICE CONTROL	EXECUTION CERTAINTY
STOP-LOSS ORDER	↓ Low (after trigger)	↑ High (after trigger)

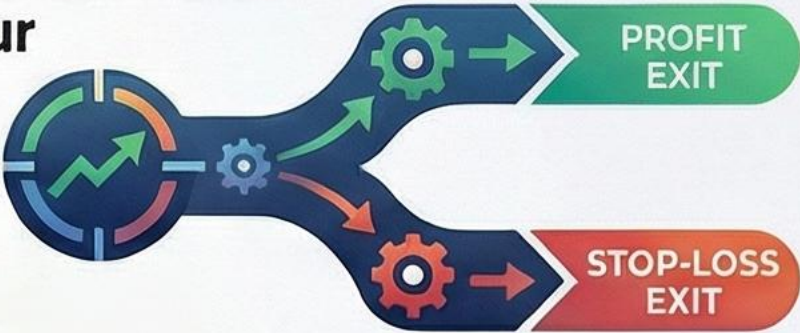
	PRICE CONTROL	EXECUTION CERTAINTY
STOP-LOSS ORDER	↓ Low (after trigger)	↑ High (after trigger)



ADVANCED & ALGORITHMIC EXECUTION

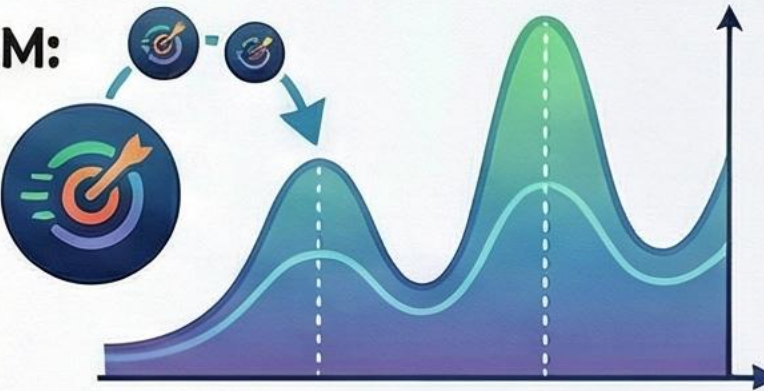
BRACKET & OCO ORDERS: Automating Your Strategy

Automatically places profit-taking and stop-loss exits once a position is opened.



VWAP ALGORITHM: Trading with the Market's Flow

Breaks up large orders, executing more when market volume is naturally high.

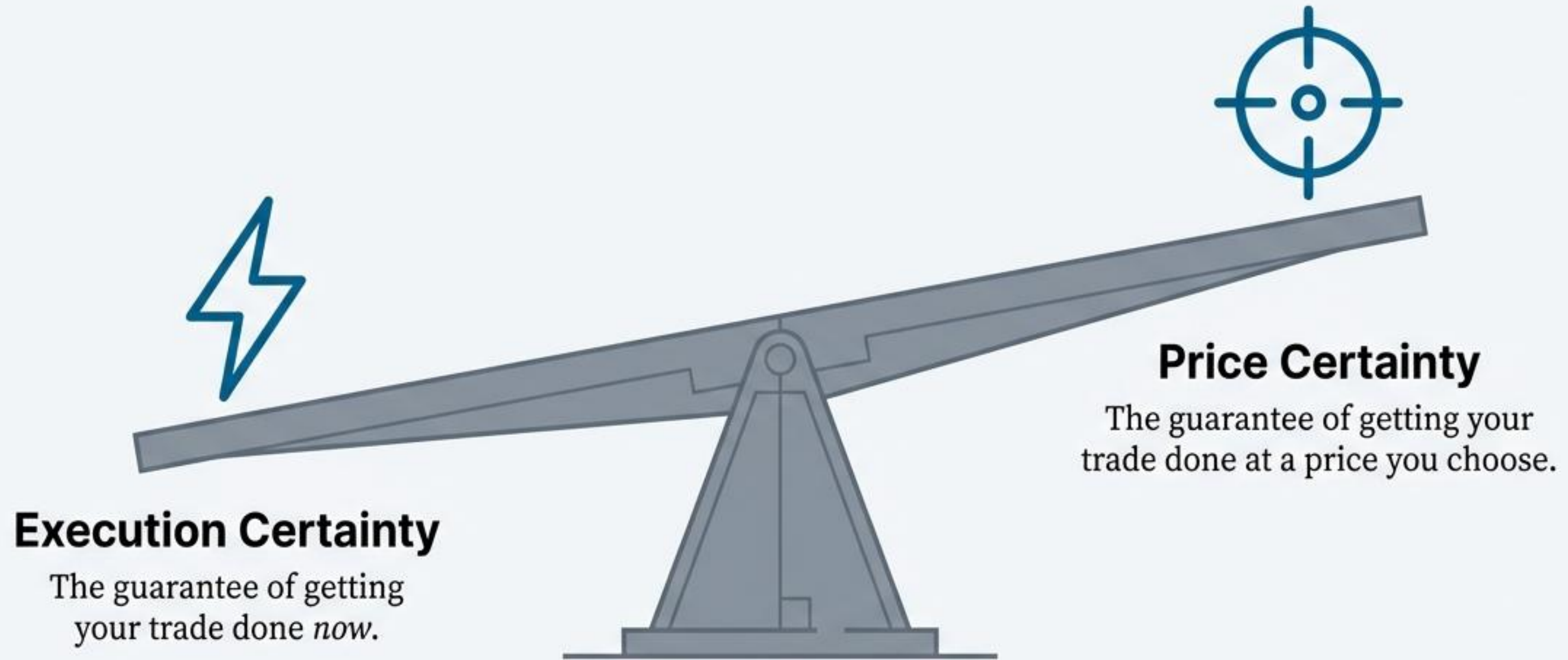


ICEBERG ORDER: Hiding Your Intentions

Reveals only a small fraction of a large order to avoid spooking the market.



Every Order Manages a Single, Core Trade-Off



The choice of order type is fundamentally a decision about where to position yourself on this spectrum. Do you prioritise speed or price? Every tool in this toolkit is designed to navigate this trade-off.

The Two Fundamental Building Blocks: Market vs. Limit Orders

Market vs. Limit Orders

Market Order

Core Function:

An instruction to buy or sell immediately at the best available market price.

Key Principle:

You are buying **Execution Certainty**.
You sacrifice price control to guarantee the trade is filled.

Certainty Trade-Off

- Execution Certainty: Virtually **Guaranteed**
- Price Certainty: Low

Limit Order

Core Function:

An instruction to buy or sell only at a specified price or better.

Key Principle:

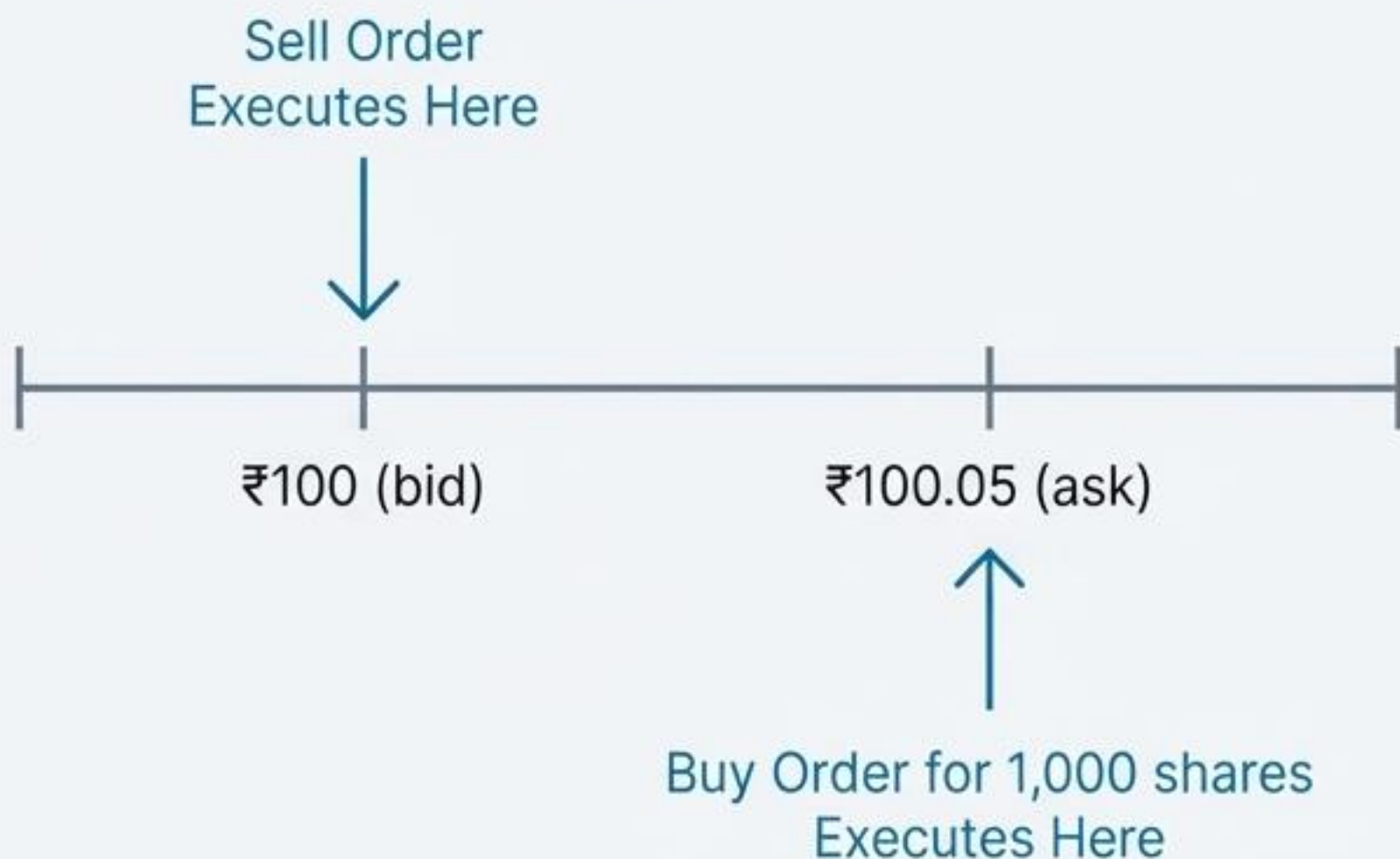
You are buying **Price Certainty**. You sacrifice execution speed and certainty to guarantee your price.

Certainty Trade-Off

- Execution Certainty: Not Guaranteed
- Price Certainty: **Complete Control**

The Market Order: Your Tool for Immediate Execution

Mechanics



- The order “takes liquidity” from the market.

Strategic Breakdown



Advantages

- Guaranteed and immediate execution.
- Simplicity; the most basic instruction.



Disadvantages

- Subject to significant slippage, especially in volatile markets.
- Can have major market impact with large orders.
- You receive the worst possible price at that moment (the bid-ask spread).



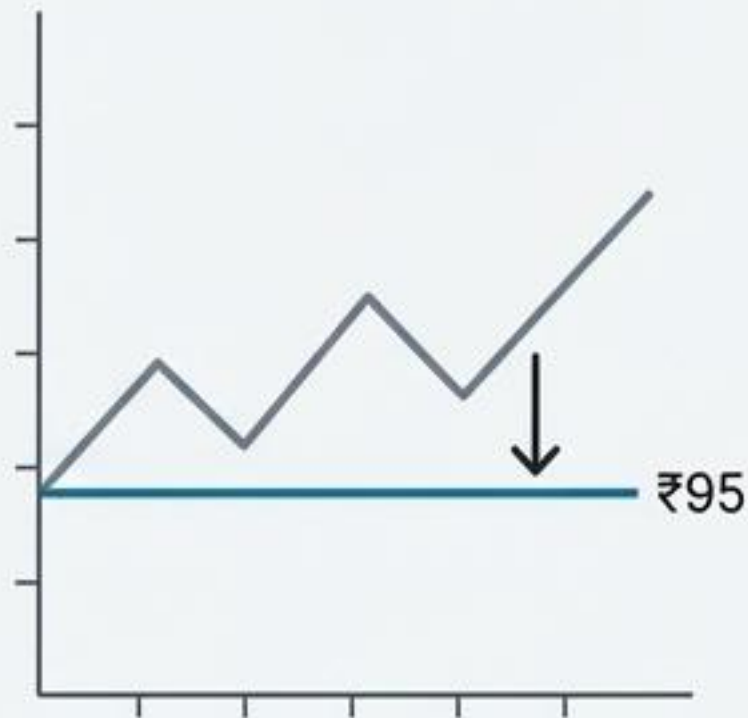
Best Use Cases

- Time-critical exits, such as closing a rapidly losing position.
- Executing small orders in highly liquid markets where the spread is minimal.
- When certainty of execution is far more important than the exact price.

The Limit Order: Your Tool for **Absolute Price Control**

Mechanics

Buy Limit



Buy Limit: Sets the *maximum* price you will pay. Order executes only if the market price falls to or below your limit. (e.g., 'Buy 100 shares at ₹95 or less').

Sell Limit



Sell Limit: Sets the *minimum* price you will accept. Order executes only if the market price rises to or above your limit. (e.g., 'Sell 100 shares at ₹105 or higher').

Strategic Breakdown



Advantages

- Complete price control; execution price will never be worse than your limit.
- From a quant perspective, there is *zero slippage* upon execution.
- Ideal for strategies requiring precise entry/exit levels.



Disadvantages

- No guarantee of execution; the price may never reach your specified level.
- Risk of missing a trade entirely if the market moves away from your price.



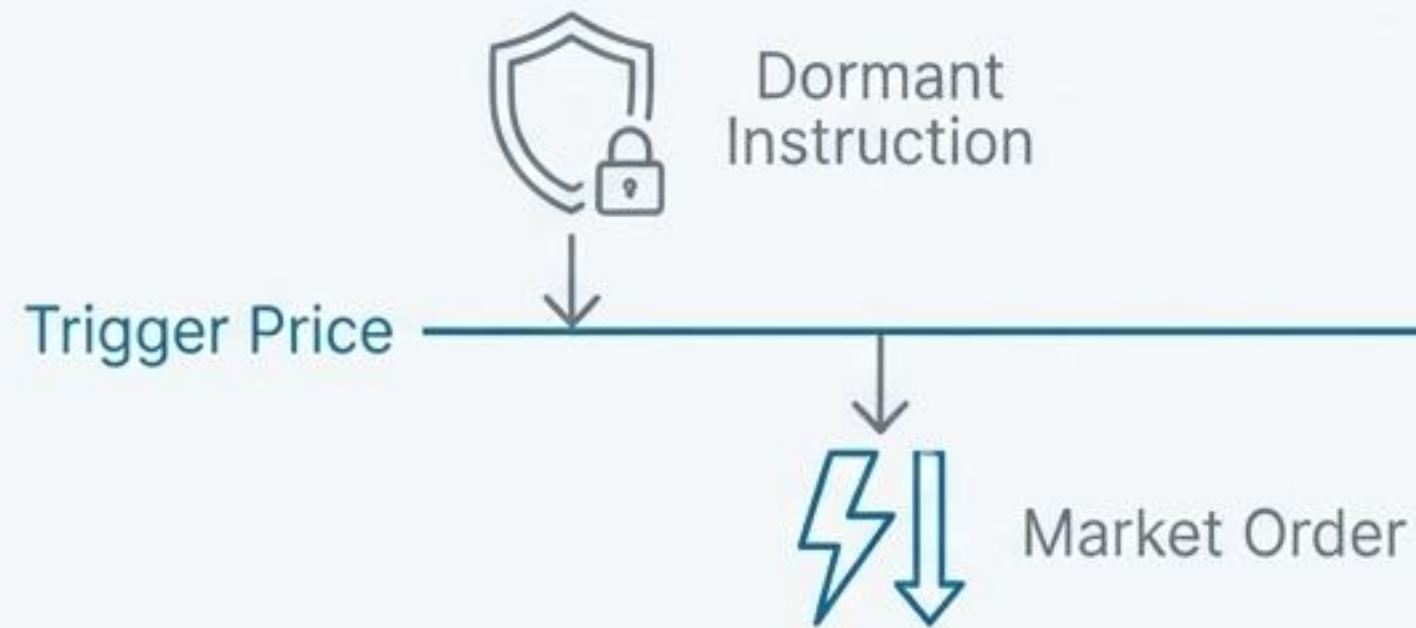
Best Use Cases

- Mean reversion strategies waiting for a price to return to a specific level.
- Trading based on support and resistance levels.
- Pairs trading, which requires specific relative prices for execution.

Building in Protection: Using Stop Orders for Risk Management

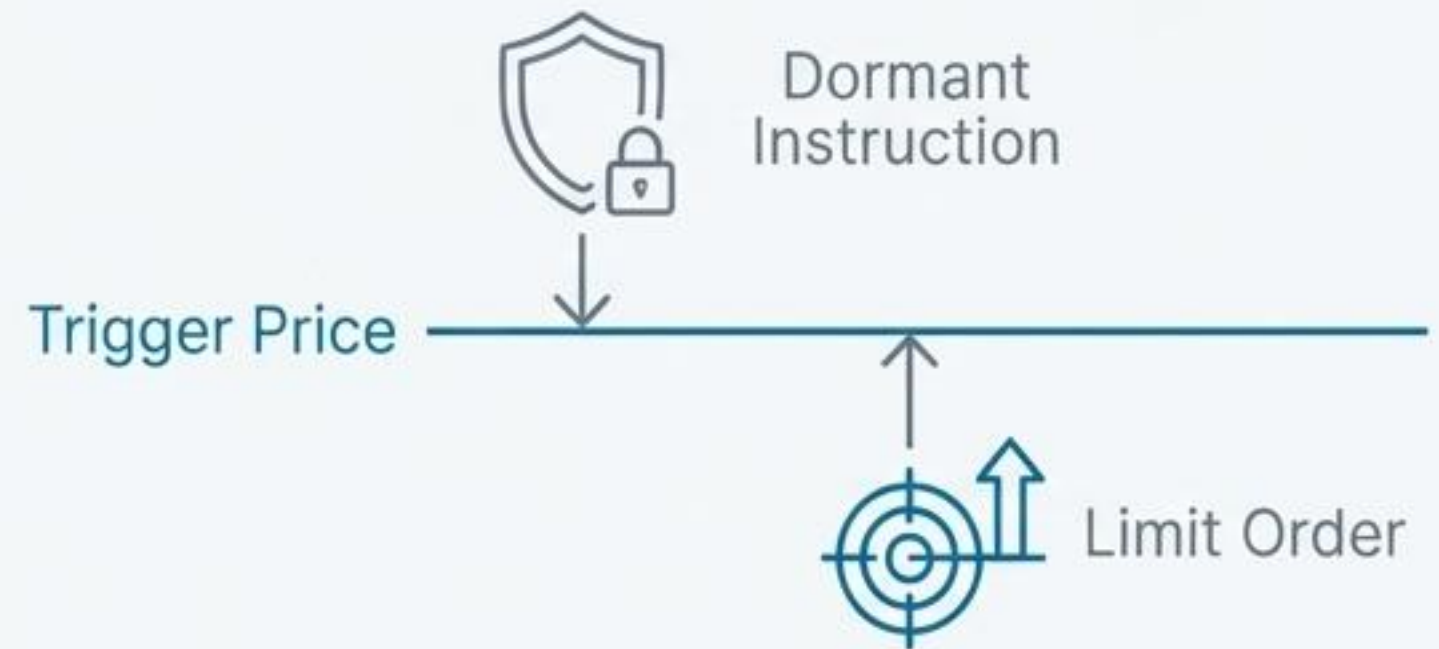
A stop order is a dormant instruction that becomes active only when a specific “trigger” price is reached. Its primary function is to **enforce discipline and automate exits**.

1. Stop-Loss Market Order ('Stop Order')



- **Mechanics:** When the trigger price is hit, it converts into a **Market Order**.
- **Goal:** Guarantees an exit from a losing position.
- **Trade-Off:** Sacrifices price control for execution certainty once triggered.

2. Stop-Limit Order



- **Mechanics:** When the trigger price is hit, it converts into a **Limit Order**.
- **Goal:** Exits a losing position but only within a pre-defined price range.
- **Trade-Off:** Balances the need for an exit with price protection, but risks non-execution if the price gaps past the limit.

Stop-Loss vs. Stop-Limit: The Critical Impact of Gap Risk

You own a stock at ₹100. You want to limit your loss to ₹5 per share.

Using a Stop-Loss at ₹95



Setup: A single sell stop-loss order is placed with a trigger price of ₹95.

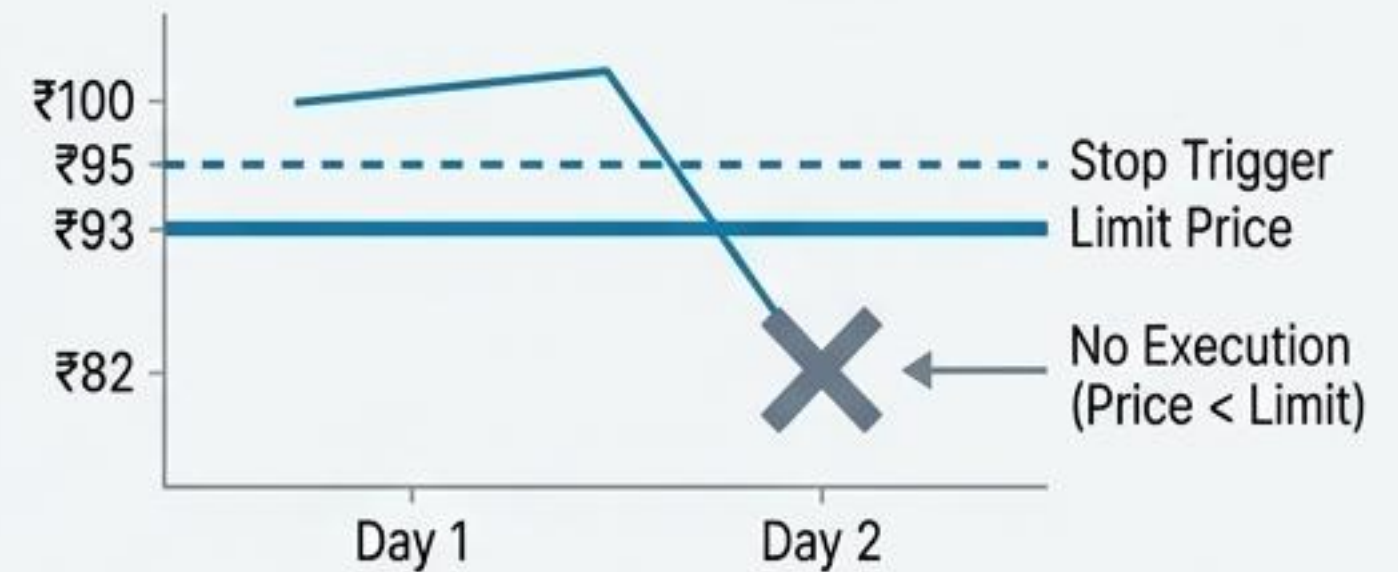
Gap Risk Event: The company releases bad news overnight. The stock opens the next day at ₹82.

Outcome:

- 📈 The trigger price of ₹95 is breached at the open.
- 📉 The order immediately converts to a **market order**.
- 🕒 It executes at the best available price, which is ~₹82.

Result: Your intended ₹5 loss becomes an actual ₹18 loss. Execution was certain, but the price was catastrophic.

Using a Stop-Limit with Stop at ₹95, Limit at ₹93



Setup: A stop-limit order is placed with a stop price of ₹95 and a limit price of ₹93.

Gap Risk Event: The same overnight news causes the stock to open at ₹82.

Outcome:

- 📈 The trigger price of ₹95 is breached.
- 📉 The order converts to a **limit order** to sell at ₹93 or better.
- 🕒 Since the market price is ₹82 (which is worse than ₹93), the limit order **does not execute**.

Result: You are still holding the losing position. You avoided a catastrophic sale price, but you did not exit the trade.

Full Automation: Bracketing a Position with OCO and Bracket Orders

These orders link multiple instructions together, allowing you to define your profit target and your maximum loss simultaneously. When one is executed, the other is automatically cancelled.

1. One-Cancels-Other (OCO) Order

Pairs two independent conditional orders. Typically, a profit-taking Limit Order and a protective Stop-Loss Order.



Benefit:

Prevents accidental duplicate trades or unwanted open orders after a position is closed.

2. Bracket Order

A comprehensive three-part order that combines an entry order with a pre-packaged OCO exit.



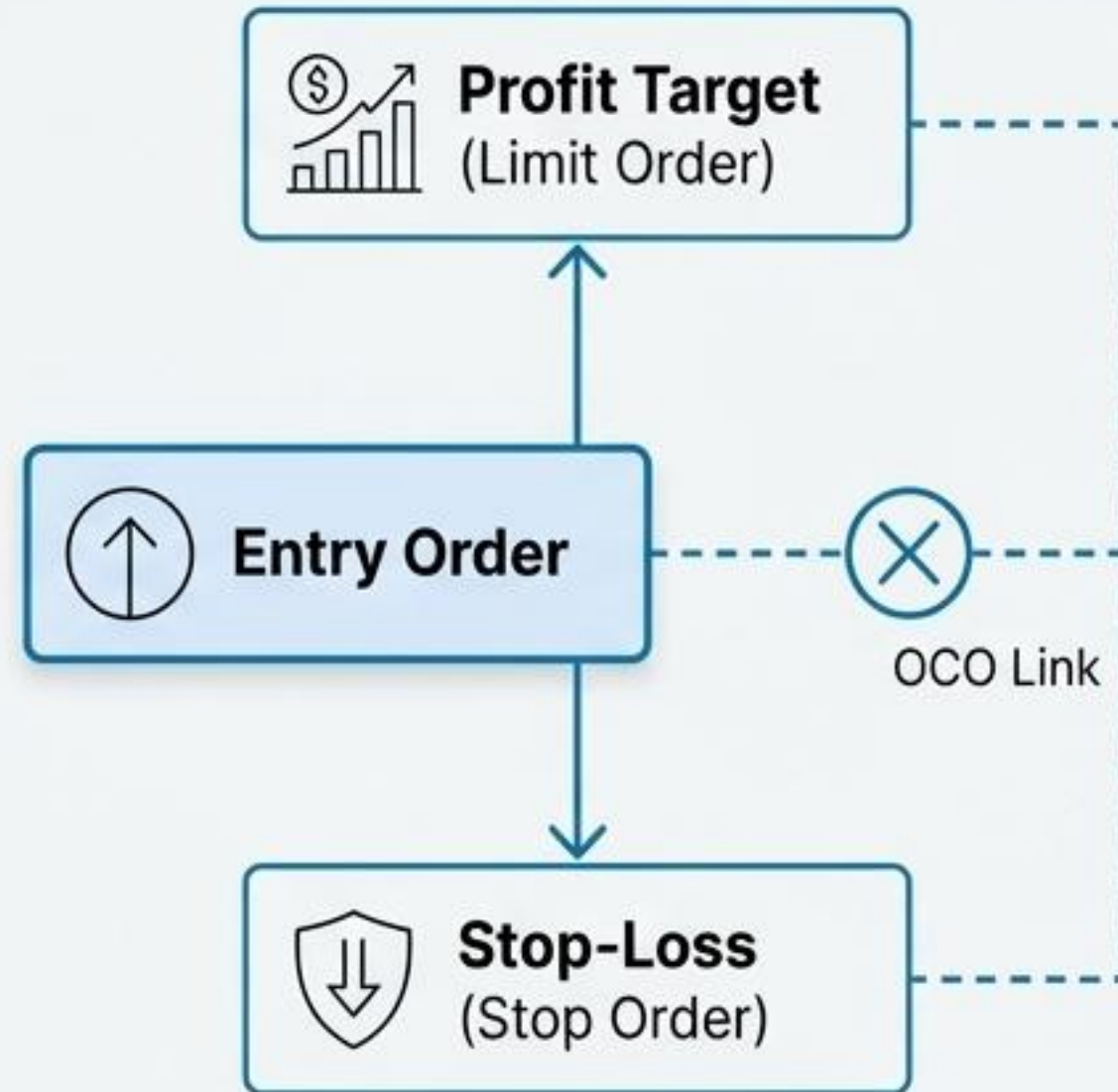
Benefit:

Complete entry-to-exit automation in a single instruction.

Anatomy of a Bracket Order

Example: Bracket order for INFY stock, currently at ₹3,665.

- **Entry:** Buy 100 shares at ₹3,665.
- **Profit Target:** Sell 100 shares at ₹3,750 (a Limit order).
- **Stop-Loss:** Sell 100 shares at ₹3,575 (a Stop order).



Order Lifecycle

1. The entry order executes at ₹3,665, establishing the position.
2. The two exit orders (profit target and stop-loss) are immediately activated as an OCO pair.
3. If the price rises to ₹3,750, the limit order fills, closing the position for a profit, and the stop-loss is automatically cancelled.
4. If the price falls to ₹3,575, the stop order triggers, closing the position for a loss, and the profit-target limit order is automatically cancelled.

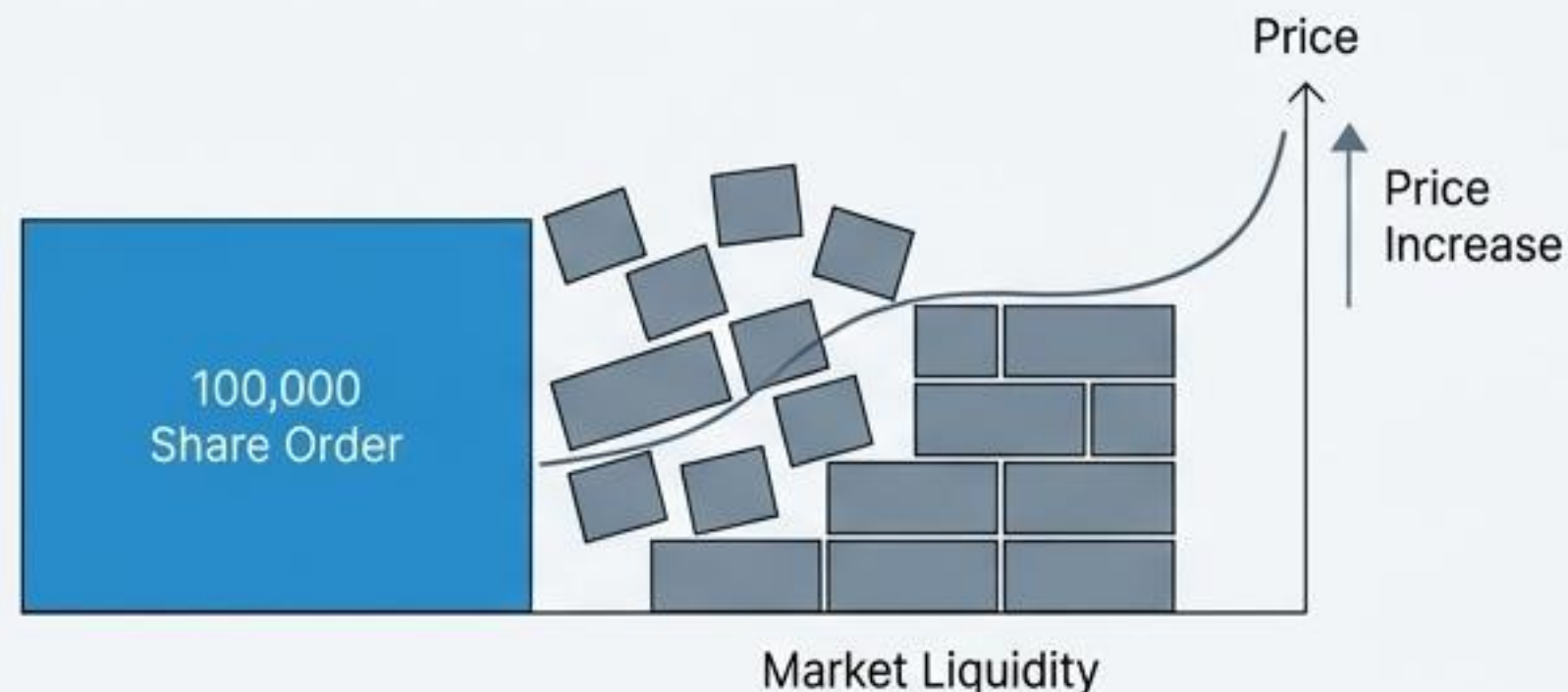
Advanced Feature: The Trailing Stop-Loss

A dynamic stop that automatically adjusts as the price moves in your favour.

Example: Set a trailing stop of ₹5 below the highest price. If the stock, bought at ₹100, rises to ₹110, the stop automatically moves from ₹95 up to ₹105, locking in a guaranteed ₹5 profit.

The Challenge of Scale: Why Large Orders Require a Different Toolkit

The Problem: Market Impact



Executing a large order (e.g., 100,000 shares) as a single market or limit order is self-defeating.



- **Price Slippage:** The price moves significantly against you as you consume available liquidity. Your average execution price is far worse than the price you saw initially.



- **Signalling:** Your large order is visible to the entire market, signalling your intentions and inviting others to trade against you (front-running).

The Solution: Algorithmic Order Slicing



Break the large "parent" order into many smaller "child" orders.



- Execute these slices strategically over time and/or across different venues.



- **Goal:** To participate in the market's natural liquidity flow rather than creating a disruptive wave.

Execution Algorithms: Time-Based (TWAP) vs. Volume-Based (VWAP)



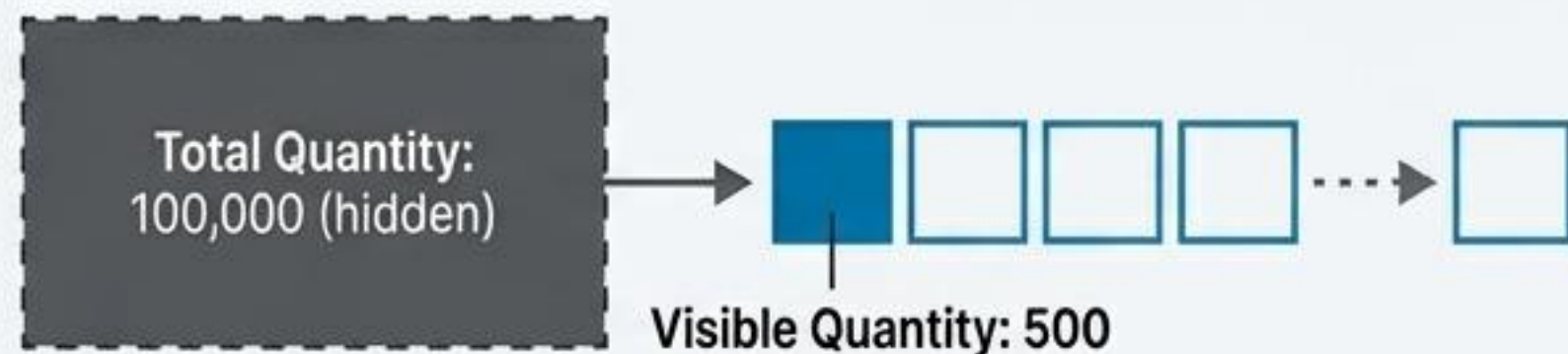
Feature	TWAP (Time-Weighted Average Price)	VWAP (Volume-Weighted Average Price)
Logic	Divides the order into equal sizes executed at regular time intervals.	Allocates execution proportional to historical or expected volume patterns.
Pacing	Predictable, but 'dumb'. Ignores market activity.	Adaptive. Trades more when the market is more active.
Advantage	Simple to implement and understand.	Aligns with natural liquidity, often achieving better prices and lower impact.
Disadvantage	Can execute unfavourably during low-volume periods.	Requires historical volume data; more complex to implement.

Advanced Algorithms: Hiding Intent and Minimising Shortfall

Iceberg Orders: Hiding True Size



Mechanics



The order reveals only a small, visible 'slice' to the market. When that slice is filled, the next slice automatically appears.

Primary Goal

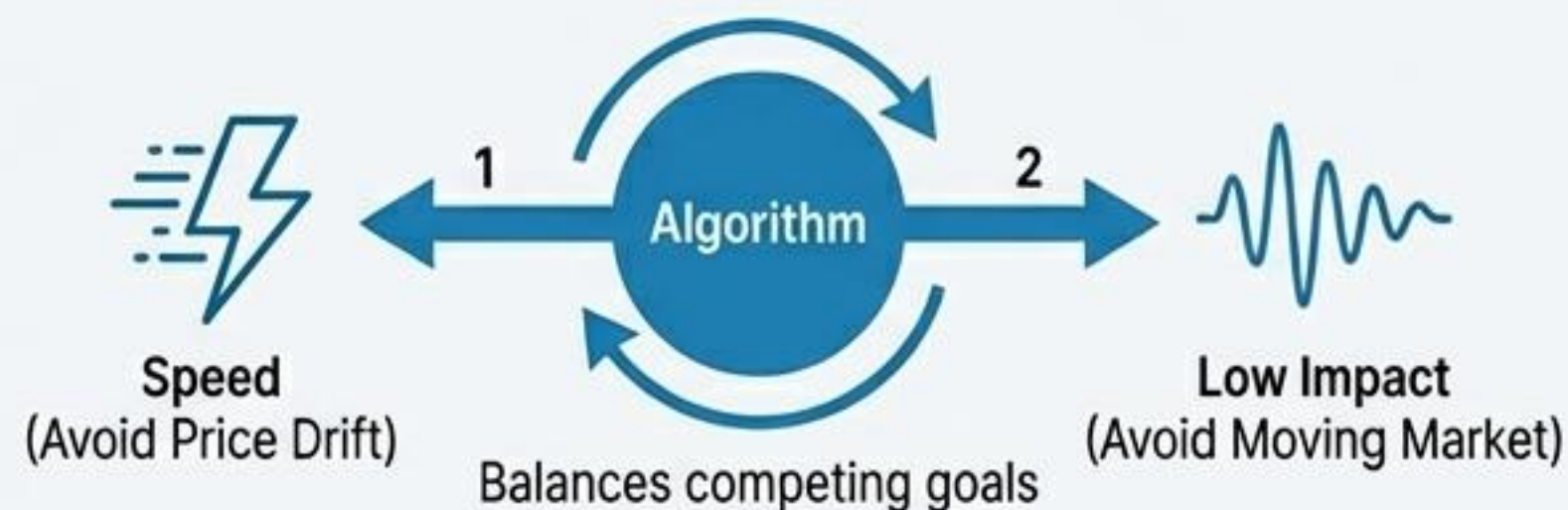
Minimise front-running risk and market impact by concealing the true size of your trading intention. Best for large orders in less-liquid markets.

Implementation Shortfall: Minimising Cost vs. Arrival Price

Core Concept

Aims to minimise the 'shortfall,' or the difference between the final average execution price and the market price at the moment the decision to trade was made (the 'arrival price').

Adaptive Logic



It becomes more aggressive when market conditions are favourable and more passive when they are not.

Primary Goal

For when minimising deviation from the decision price is the most critical benchmark.

Specifying an Order's Lifespan: Time-in-Force

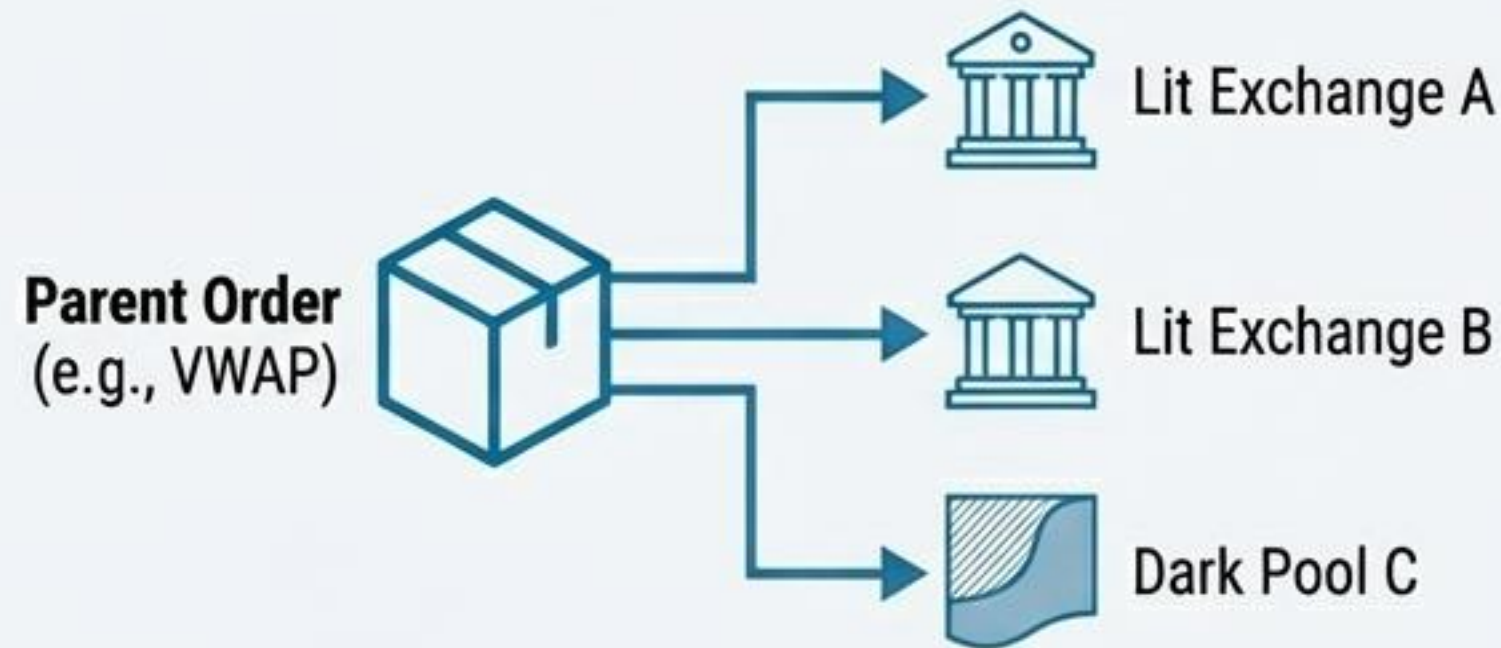
Time-in-Force (TIF) instructions tell the exchange how long an order should remain active before being cancelled.

TIF Instruction	Description	Use Case
DAY	The order is active only for the current trading day. Automatically cancelled at market close.	Default for most systems. Intraday strategies where you want a clean slate each morning.
GTC (Good-Till-Cancelled)	The order remains active across multiple trading days until it is either filled or manually cancelled.	Patient entry/exit points at key support/resistance levels that you are willing to wait days or weeks for.
IOC (Immediate-or-Cancel)	The order must be executed immediately for whatever quantity is available. Any unfilled portion is cancelled. Partial fills are allowed.	Testing for available liquidity at a specific price point without leaving a resting order on the book.
FOK (Fill-or-Kill)	The entire order must be filled immediately. If the full size is not available, the entire order is cancelled. No partial fills.	All-or-nothing situations like pairs trading or arbitrage where both legs must execute simultaneously and in full.

The Intelligence Layer: Smart Routing and Dynamic Pricing

1. Smart Order Routing (SOR)

Algorithms that automatically route order slices to the optimal execution venue (Lit Exchange A, Dark Pool B, etc.) to achieve the best results.



Objectives

- ✓ Source the best available price across all connected markets.
- ✓ Minimise transaction costs and market impact.
- ✓ Navigate complex regulatory requirements.

SOR acts as the 'logistics brain' for execution algorithms like VWAP, deciding *where* to send each child order.

2. Pegged Orders

An order whose price is not fixed, but instead automatically adjusts based on a reference point, such as the best bid, best ask, or midpoint of the spread.

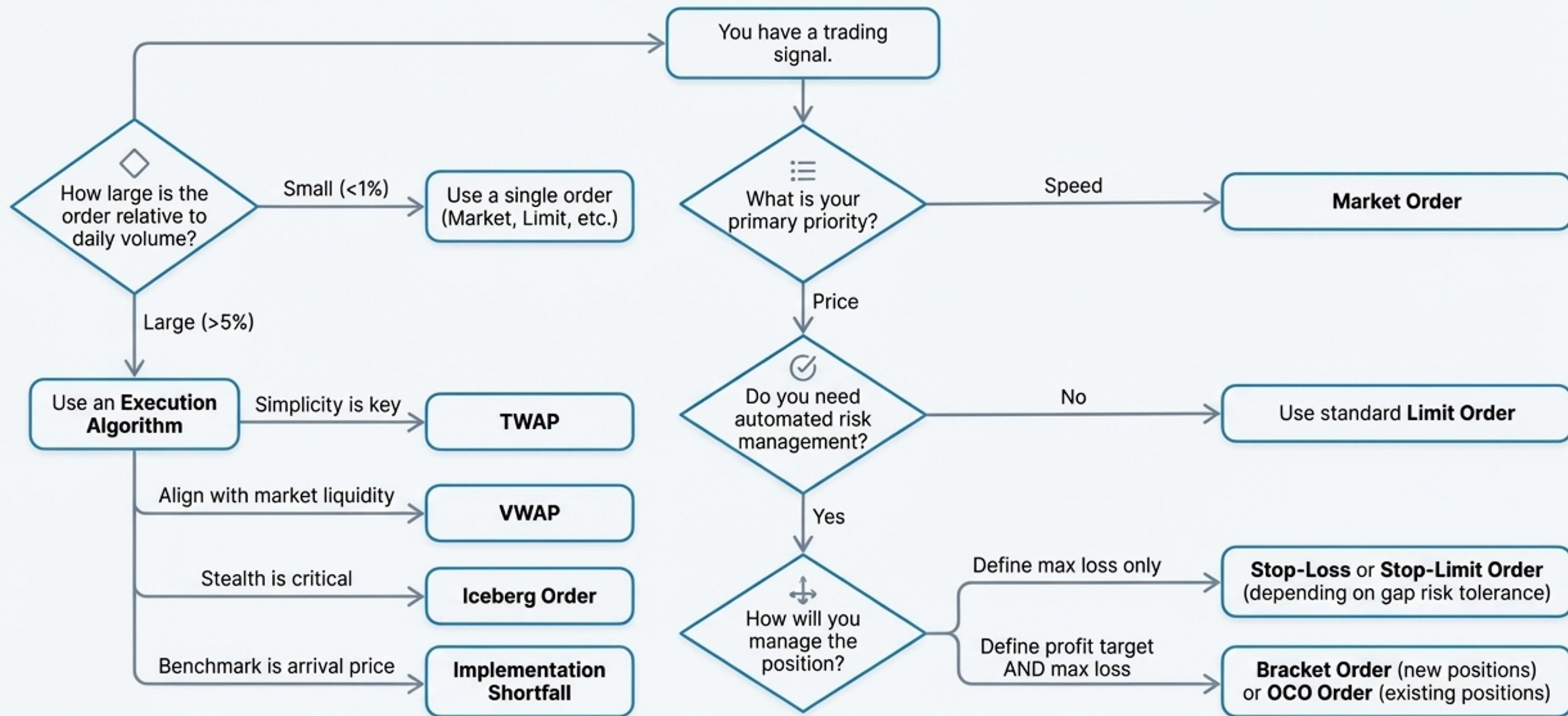


Example

A 'Pegged to Midpoint' order will always rest precisely in the middle of the bid-ask spread, moving dynamically as the spread shifts.

Allows for passive participation in the market, keeping an order competitive without requiring constant manual updates. Essential for market-making strategies.

The Quant's Order Selection Framework



Mastering order types is the critical interface between trading logic and market execution. This framework helps translate your strategy into precise, executable instructions.